CONSERVATION FUNDING ALLOCATION PROJECT

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Introduction

The goal of this project is to identify regions in Africa where conservation funding has the most impact, based on the data below:

- Biodiversity data – species recorded, species type (Plant or Mammal)
- Conservation funding in millions (USD)
- Biodiversity intactness index – percentage of original species that remain and are abundance in an area despite human impact.

This data is cleaned, analyzed, outputs created, and recommendations made.

Data Cleaning and Preparation

This involves identifying and correcting errors, handling missing values, standardizing formats and removing duplicates. This is to ensure accuracy and consistency when analyzing data to derive meaningful information.

The first step in this project was to load both the biodiversity excel file and the conservation funding csv file. The biodiversity excel file had 11 sheets that were combined into one dataframe.

The next step was to check for spelling errors in the biodiversity and conservation funding dataset. Using python language in jupyter lab, fuzzywuzzy library was used that compares string matches in the dataset.

For the biodiversity dataset, the following spelling errors were identified and corrected:

```
Spelling Errors Detected: {'Congo': 'Republic of Congo', 'Djiboutg': 'Djibouti', 'Eswatinb':
'Eswatini', 'Ethiopir': 'Ethiopia', 'Gambil': 'Gambia', 'Gambie': 'Gambia', 'Ghanh': 'Ghan
a', 'Guinee': 'Guinea', 'Guineu': 'Guinea', 'Ivory Coasj': 'Ivory Coast', 'Ivory Coase': 'Iv
ory Coast', 'Mauritaniw': 'Mauritania', 'Mauritiud': 'Mauritius', 'Nigew': 'Niger', 'Sao Tom
e and Principr': 'Sao Tome and Principe', 'Senegaz': 'Senegal', 'Sierra Leonx': 'Sierra Leon
e', 'Sudaa': 'Sudan', 'Tanzanir': 'Tanzania', 'Ugands': 'Uganda', 'Ugandb': 'Uganda'}
```

For the conservation dataset, the following spelling errors were identified and corrected:

```
Spelling Errors Detected: {'Congo': 'Republic of Congo', 'Ethiopix': 'Ethiopia'}
```

Using python in the jupyter lab, a pandas profiling library was used to automate the process of exploratory data analysis. It is useful for identifying missing values, duplicates and statistical distribution of data. For this project, a report was generated on the biodiversity and conservation funding data and the results are as follows:

- Biodiversity data

From the report: Biodiversity_data_statistics_report.html, this data is categorical and text. There are 1936 species in 53 African countries based on the data provided. However, there are 6 duplicates. These are then removed from the dataframe and a total of 1930 observations remain indicating there are 1930 species in Africa categorized as either plant or mammal.

- Conservation funding

From the report:Conservation_fund_statistics_report.html, the data is both numeric and text. There are 53 observations in the data indicating that there are 53 countries in Africa with funding data available. However, there is one country in Africa in the dataset with missing funding data. This is then replaced by 0 to make it easier to do the analysis. There is also an African with a negative value for funding. An assumption was made that value was outlier and converted to a positive value.

Some countries in both datasets had inconsistent naming such as Ivory Coast which was renamed to Côte d'Ivoire match naming in both conservation funding and biodiversity data frame.

The shapefile containing African countries polygon was merged with both cleaned conservation funding and biodiversity dataframe. Duplicates were checked in the dataframe and were found to be none. Missing values in the shapefile were also checked and replaced with 0. Relevant columns such as “ADM0_CODE” was renamed to Country for consistency with the other dataframes. Countries with inconsistent names with other the dataframes e.g. Cape Verde and Swaziland were renamed to Cabo Verde and Eswatini respectively.

To obtain Biodiversity Intactness Index for each country, A zonal Statistics (Mean) was run on both the raster file and the merged dataframe with the Countries’ geometry, species name, species type and conservation funding. The mean calculated the average of the index value around the geometry of each African country and areas with no index value on the raster assigned a value of 0 instead of No data.

The final shapefile with country name, species name, species type, funding allocation, geometry, species count, species type count and biodiversity intactness index was exported.

Data Visualization

For the cleaned conservation dataset, the following statistics were generated:

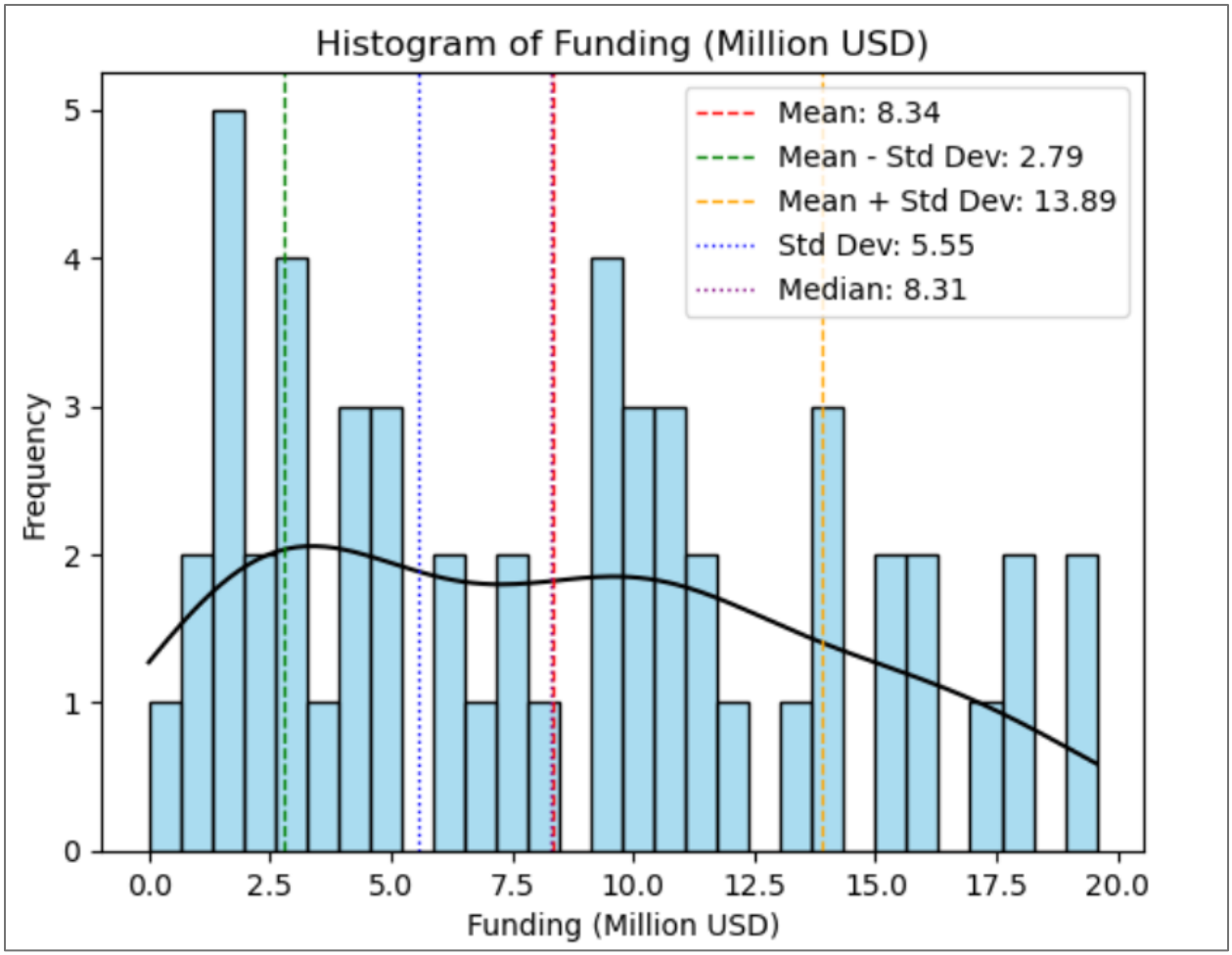

Figure 1: Conservation funding histogram

From the histogram above, funding for biodiversity in Africa ranges from 0 to 20 million. The mean is 8.34 indicating averaging funding in Africa for conservation is 8.34 million. The histogram shows that most funding cluster between 2.79 million USD (mean minus one standard deviation) and 13.89 million USD (mean plus one standard deviation) with few countries receiving conservation funding beyond this range. It also shows that the data distribution is right skewed (positively skewed). This shows that most countries in Africa most countries in Africa receive low funding while a few countries receive large amounts of funding.

For Biodiversity, it was merged with the conservation dataset into one data frame. and the following statistics were generated:

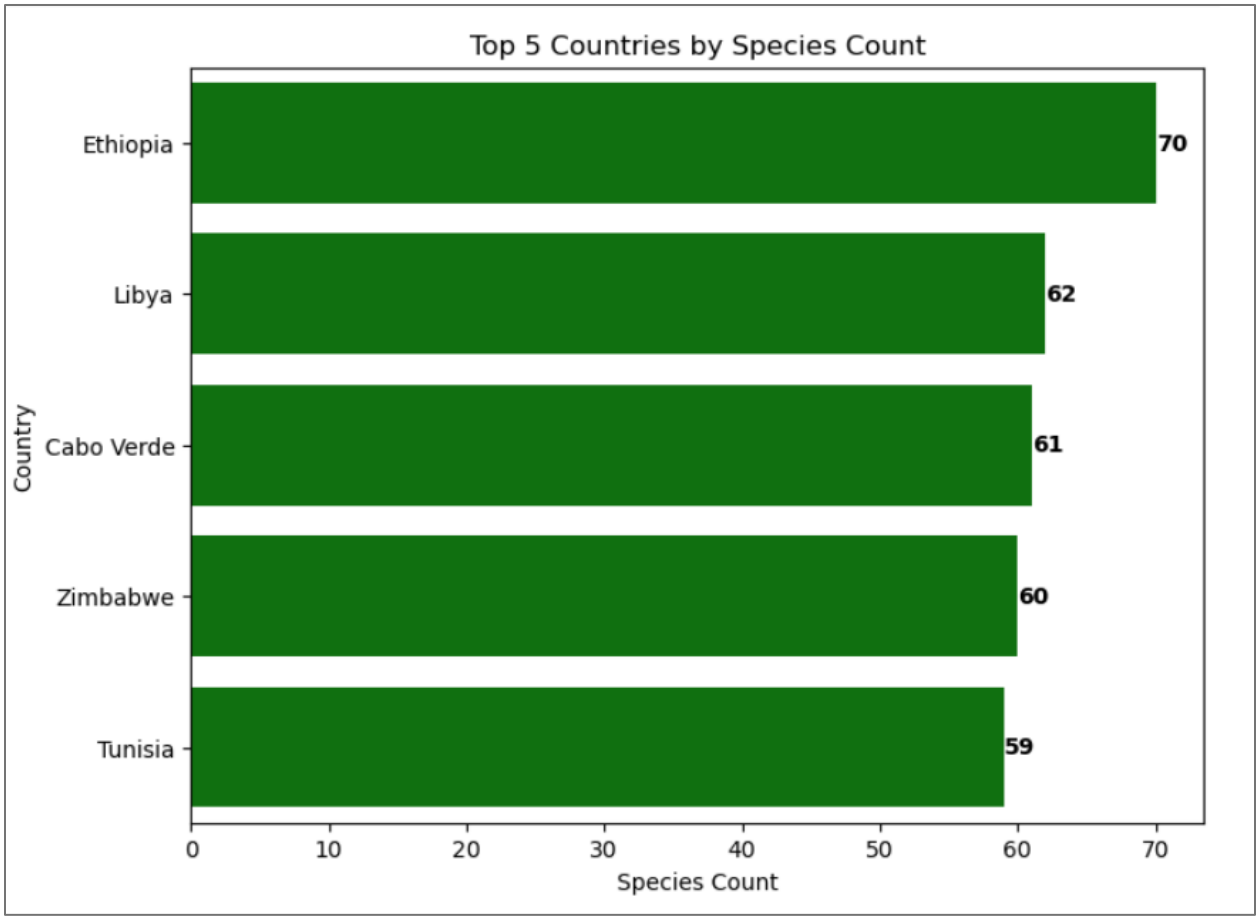

Figure 2: Top 5 countries on species count

In the bar graph above, Ethiopia has the highest number of species (70). This is followed by Libya (62 species), Cabo Verde (61 species), Zimbabwe (60 species) and Tunisia (11 species).

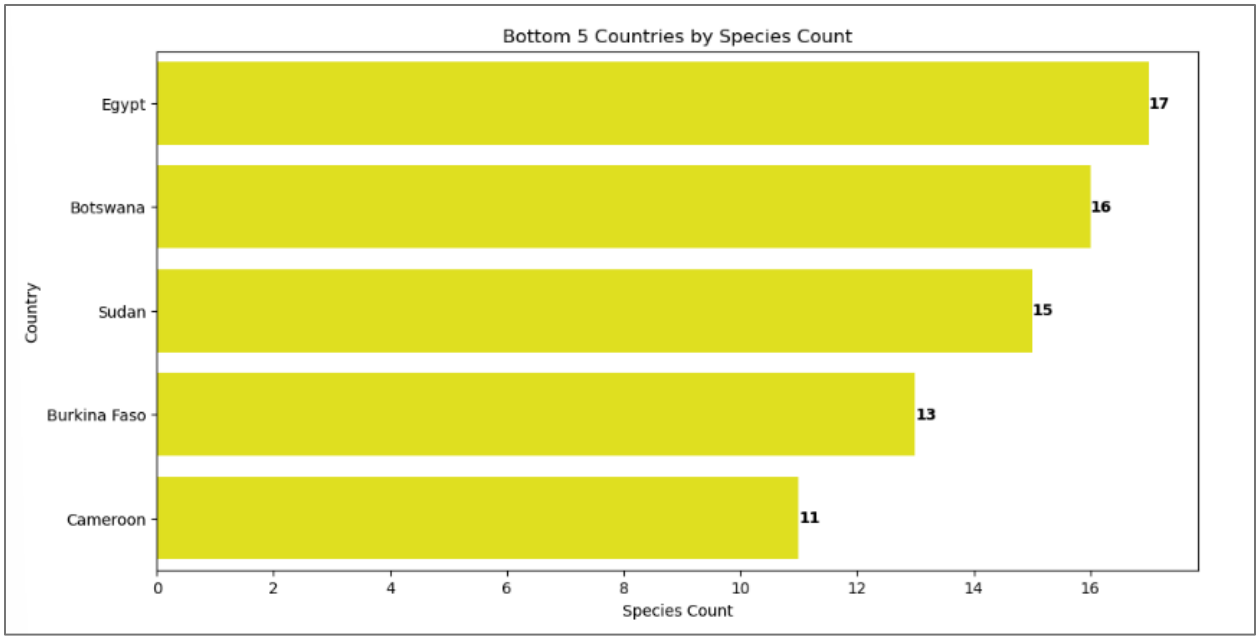

Figure 3: Bottom 5 countries on species count

In the bar graph above, Cameroon has the lowest number of species (11). This is followed by Burkina Faso (13 species), Sudan (15 species), Bostwana (16 species) and Egypt (17 species).

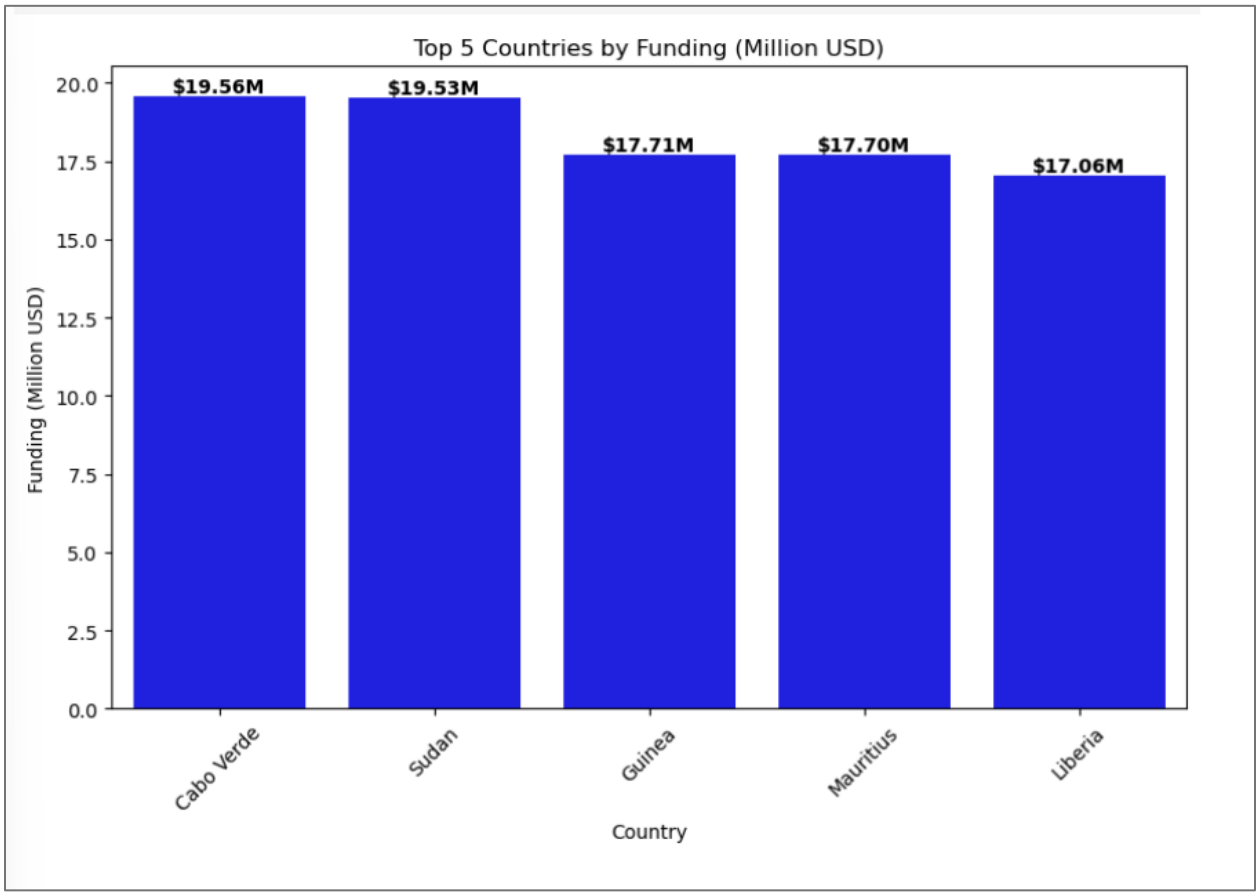

Figure 4: Top 5 countries on conservation funding

In the bar graph above, Cabo Verde had the highest amount of funding (19.56 Million USD). This is followed by Sudan (19.53 Million USD), Guinea (17.71 Million USD), Mauritius (17.70 Million USD) and Liberia (17.06 Million USD)

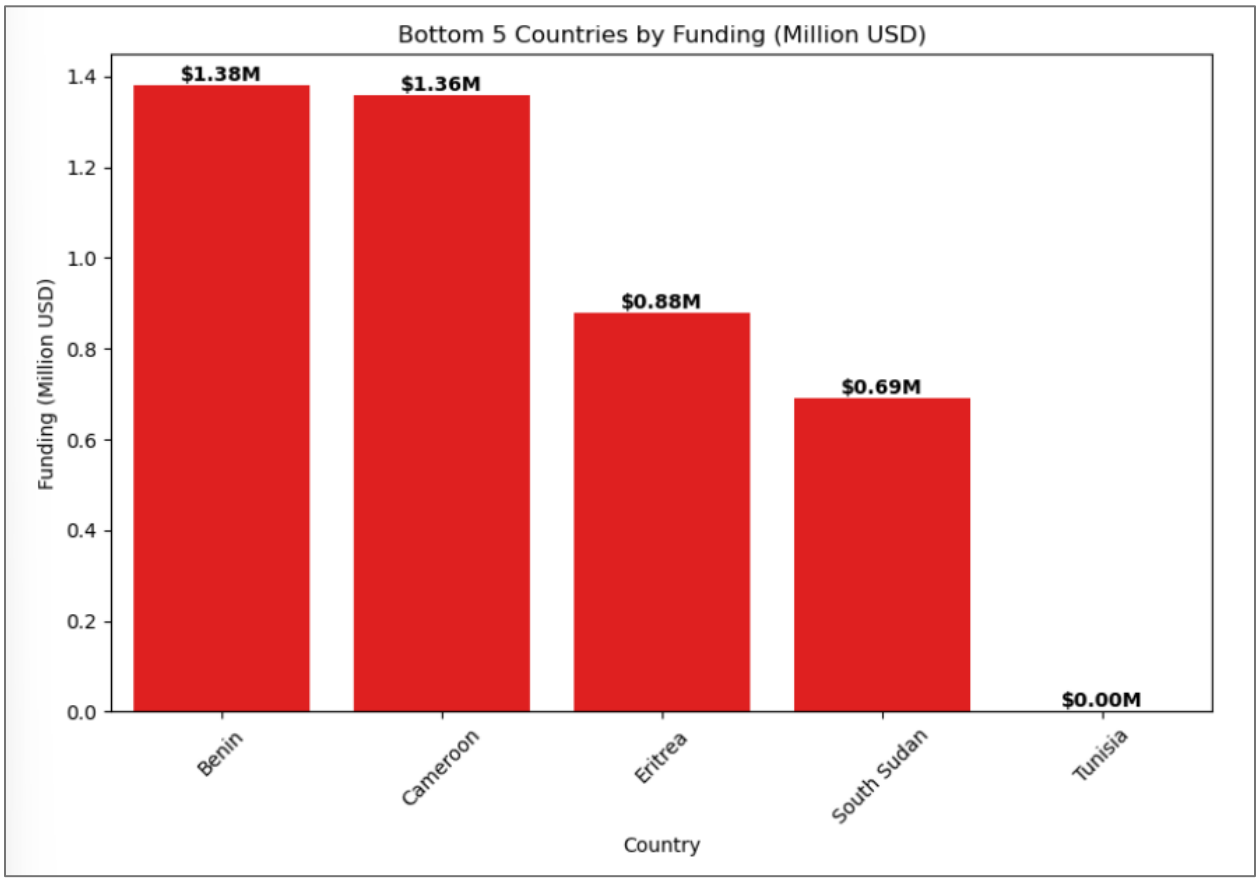

Figure 5: Bottom 5 countries on conservation funding

In the bar graph above, Tunisia had the lowest amount of funding (0 Million USD). This is followed by South Sudan (0.69 Million USD), Eritrea (0.88 Million USD), Cameroon (1.36 Million USD), and Benin (1.38 Million USD).

For the African countries shapefile, the relevant statistics was the count for column ADM0_CODE.

	ADM0_CODE
count	59.000000

Figure 6: Count on African Territories

From the figure above, The count was 59 indicating there were 59 territories in Africa.

For the Biodiversity intactness index, which is a raster dataset, the following statistics were generated:

Out[58]:

	Property	Value
0	Mean Intactness	0.756095
1	Max Intactness	0.995363
2	Min Intactness	0.116944
3	Width (px)	975
4	Height (px)	809
5	CRS	GEOGCS["WGS 84",DATUM["World Geodetic System 1...
6	Bounding Box	(-18.03817, -35.070229999999995, 52.03042, 23....

Figure 7:Biodiversity Intactness Index raster statistics

In the table above, the minimum and maximum range of values is 0.116994 to 0.995363, indicating that the biodiversity index values are in this range. The mean of Biodiversity intactness is 0.756095 indicating that the average Biodiversity Intactness Index in Africa is 0.756095. The Coordinate Reference of the raster dataset is WGS 1984.

The maps below were created using the exported conservation funding data from jupyter lab.

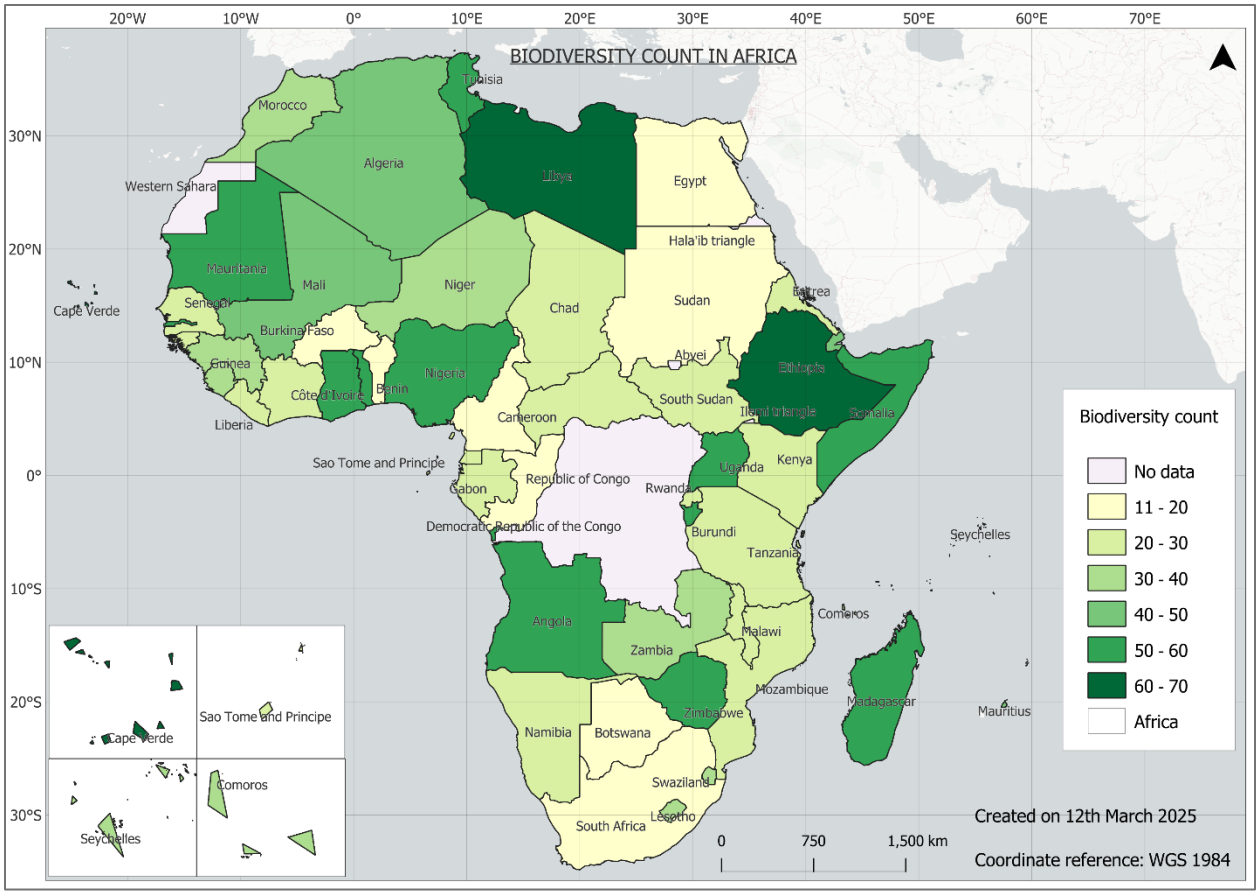

Figure 8: Africa biodiversity count map

The map above shows the Biodiversity count in the whole of Africa. In the map, two countries have no biodiversity count. In the Biodiversity dataset, there was no distinction between the Republic of Congo and Democratic Republic of Congo. Since the dataset had the country Congo, an assumption was made that the biodiversity in the dataset belongs to the Republic of Congo resulting in no data for biodiversity in DR Congo. There was no data for Western Sahara.

In the map, biodiversity is categorized into different count ranges as shown by the legend on the map with countries low species count shown in light colour ranges and countries with higher species shown in dark colour ranges of green.

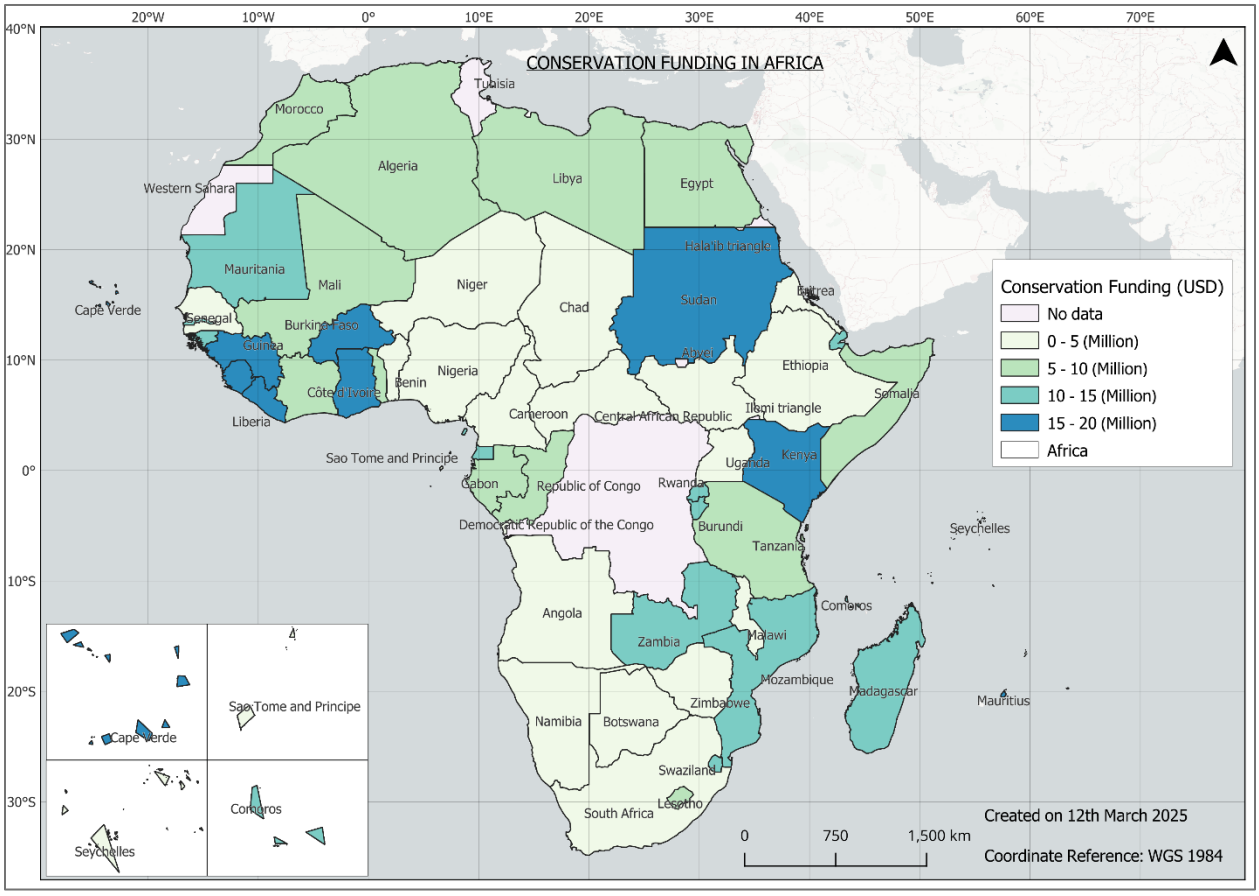

Figure 9: Conservation funding in Africa map

The map below shows Conservation in the whole of Africa. In the map, two countries have no conservation funding. In the conservation funding dataset, there was no distinction between the Republic of Congo and Democratic Republic of Congo. Since the dataset had the country Congo, an assumption was made that the conservation funding in the dataset belongs to the Republic of Congo resulting in no data for biodiversity in DR Congo. There was no data for Western Sahara.

In the map, conservation funding is categorized into different count ranges as shown by the legend on the map with countries low funding allocation in light colour ranges and countries with high funding allocation shown in dark colour ranges of blue.

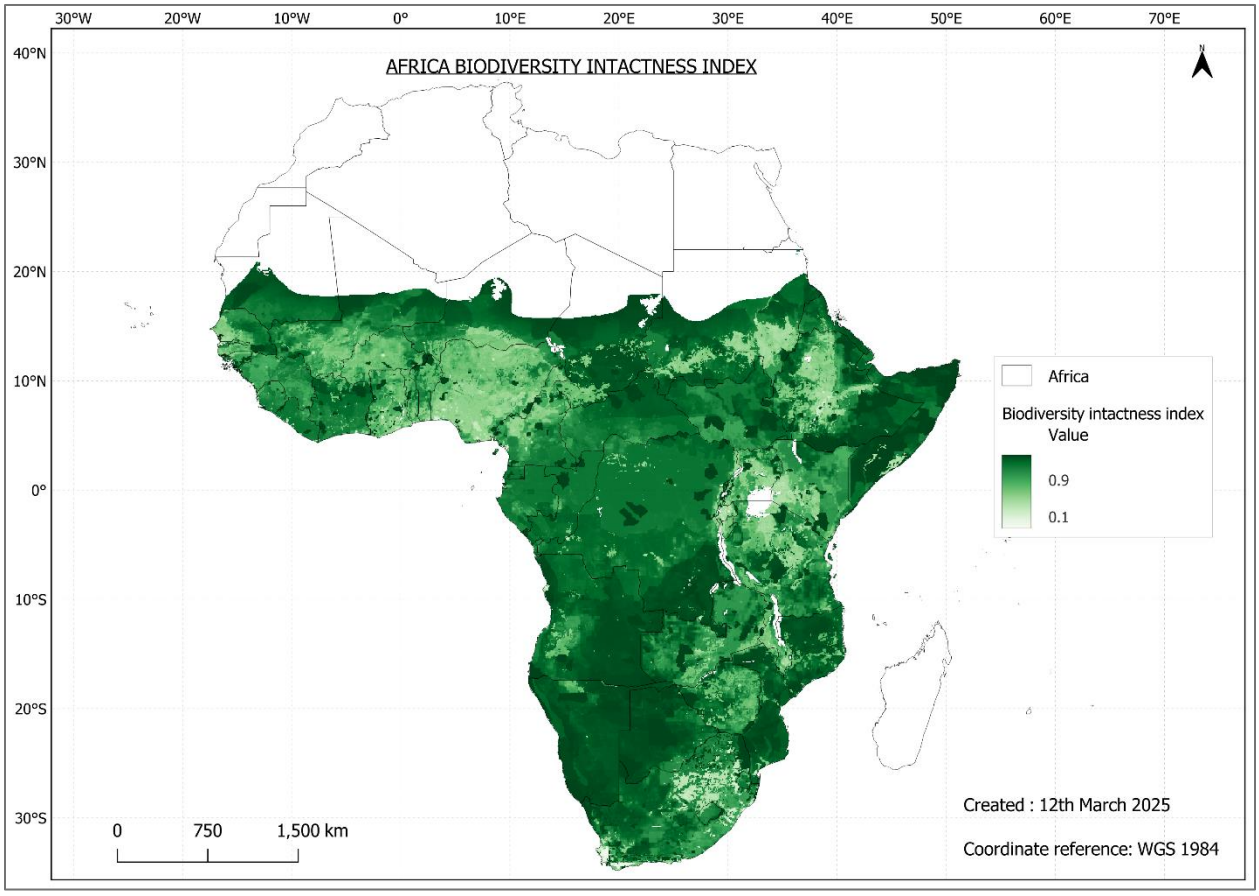

Figure 10: Africa biodiversity intactness map

The map above shows the biodiversity intactness index across Africa using a continuous green colour gradient with darker areas showing high biodiversity intactness and light areas with low biodiversity intactness index. The index values range from 0.1 to 0.9 with high values indicating high biodiversity intactness index and low values indicating low biodiversity intactness index. Countries with no biodiversity intactness index have no continuous colour gradient.

Findings

The table below shows countries in Africa with high species count and low funding. Ethiopia has the highest species count (70) but receives only 4.01 Million USD in funding. Zimbabwe (60 species) and Angola (58 species) receive moderate funding of 4.07 Million USD and 4.21 Million USD, respectively. Tunisia has 59 species but receives 0 Million USD, indicating no conservation funding despite relatively high biodiversity. Uganda and Nigeria have biodiversity counts of 58 and 57, respectively, but receive low funding of 2.86 Million USD and 2.64 Million USD.

	Country	Species_Count	Funding (Million USD)
451	Ethiopia	70	4.01
1709	Libya	62	6.25
1547	Zimbabwe	60	4.07
1854	Tunisia	59	0.00
15	Angola	58	4.21
1433	Uganda	58	2.86
1107	Nigeria	57	2.64
1378	Togo	55	7.73

Figure 11: African countries with high species and low funding

The table below shows countries high biodiversity intactness index but low funding. The Biodiversity Intactness Index measures how much of a region's original biodiversity remains intact, with values closer to 1 indicating better biodiversity preservation. However, conservation funding varies significantly across these countries as shown below. Namibia (Index: 0.871, Funding 1.52 Million USD) and Eritea (Index: 0.807, Funding 0.88 Million USD) have the highest index with low conservation funding. Bostwana (Index: 0.875, Funding 4.69 Million USD) and Angola (Index: 0.840, Funding 4.21 Million USD) have high index with moderate funding. Ethiopia (Index: 0.727, Funding: 4.01 Million USD), Zimbabwe (Index: 0.716, Funding: $4.07 Million USD) and South Africa (Index: 0.723, Funding: 2.87 Million USD) with declining biodiversity intactness index with relatively low conservation funding.

	Country	Funding (Million USD)	Biodiversity_Intactness_Index
93	Botswana	4.69	0.874608
1042	Namibia	1.52	0.871539
15	Angola	4.21	0.839816
221	Chad	2.95	0.818063
425	Eritrea	0.88	0.806772
194	Central African Republic	4.83	0.804783
284	Republic of Congo	6.50	0.801600
379	South Sudan	0.69	0.795385
521	Gabon	6.75	0.775330
1072	Niger	3.37	0.759611
451	Ethiopia	4.01	0.726827
1321	South Africa	2.87	0.722933
744	Lesotho	5.17	0.718860
1547	Zimbabwe	4.07	0.715501

Figure 12: Countries in Africa with high BI Index and low funding

Conclusion

Despite Tunisia having a high biodiversity funding, there is no data available for funding therefore difficult to determine whether it is underfunded or well supported. Uganda and Nigeria there are countries with high biodiversity count but with low funding. Ethiopia and Zimbabwe have the highest biodiversity count with moderate funding indicating that they need more funding allocation for conservation.

Countries like Namibia, Eritrea, and South Sudan have low conservation funding despite having some of the highest biodiversity index indicating a need to allocate funding to these countries. There are moderate funding levels in Angola, Bostwana, Central Africa Republic indicating they could benefit from increased conservation financing to sustain biodiversity.

Ethiopia, Zimbabwe and South Africa have low biodiversity intactness index is low, but funding is insufficient in preventing further losses.

This project highlights the need for increased conservation funding in countries where biodiversity count and intactness index are high to maintain ecological balance.

Recommendations

In the biodiversity dataset, there was no distinction between Republic of Congo and Democratic Republic of Congo for species type and species name leading to assumption in biodiversity and conservation funding analysis. Further data collection should explicitly differentiate between the two countries to ensure accurate funding allocations and biodiversity assessment. Western Sahara lacks biodiversity and conservation funding data and this should be looked into.

The right-skewed distribution of conservation funding indicates that few countries receive most conservation funding while many countries receive little funding. Funding should be re-evaluated based on each countries' conservation needs and thresholds established for biodiversity rich countries to bridge this gap.

Expanding the use of machine learning algorithms can help predict areas at risk of biodiversity loss and optimize conservation funding allocations.